

Decentralizovaná aplikace nahrazující průzkumné agentury

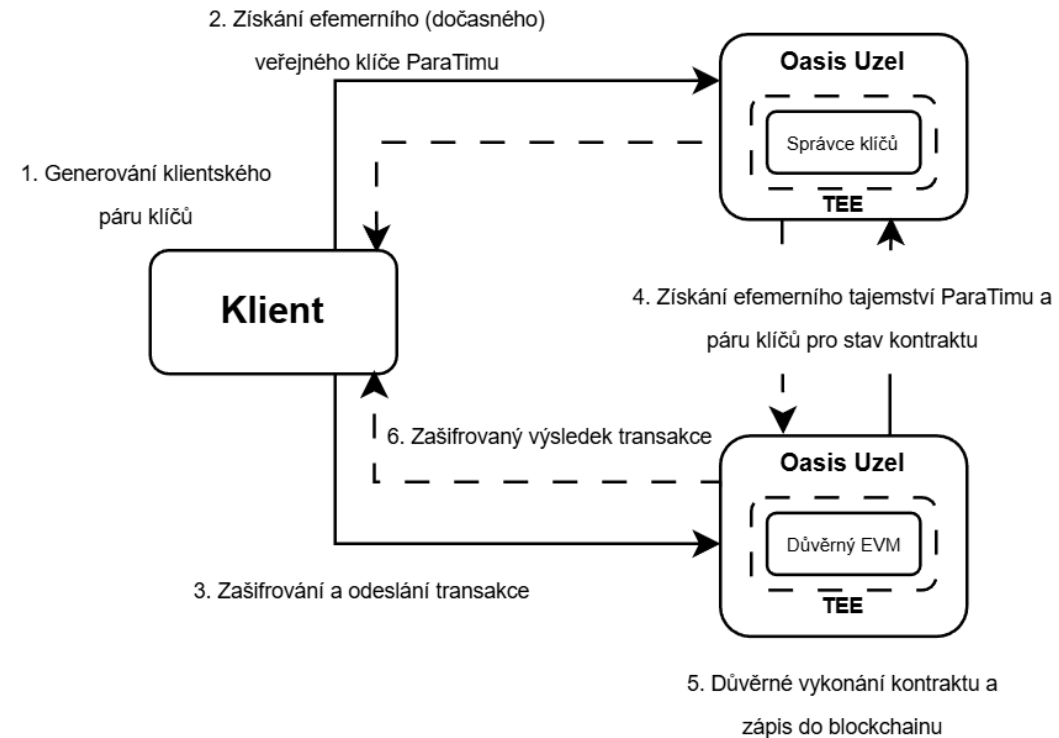
Tomáš Hanák

vedoucí: doc. Ing. Ivan Homoliak, Ph.D.



- Nedostatky centralizace
- Hrozba Sybil útoků
- Soukromí a ověřování

- Platforma Oasis Sapphire
- Důvěryhodná prováděcí prostředí (TEE)
- Důvěrný stav a ochrana citlivých dat



- Anonymita prostřednictvím protokolů zk-SNARK (Groth16)
- Vysoká efektivita a nízké transakční náklady

- Využití modelovaných dat
- Optimalizace pomocí Poseidon hashe

```
{
  "id": 101,
  "name": "Jan Novak",
  "birthDateInt": 19900515,
  "address": {
    "city": "Prague",
    "region": "Prague"
  },
  "keys": {
    "pk": ["81480...3782", "12687...72091"]
  }
}
```

- Transformace pravidel do matematických omezení
- Vnitřní logika a verifikace
- Ochrana proti opakovanému hlasování (Nullifier)

```
// Private inputs
signal input vcAttributes[8];

signal input sigR8x;
signal input sigR8y;
signal input sigS;

// Public inputs
signal input pkAx;
signal input pkAy;
signal input pkVCx;
signal input pkVCy;

signal input allowedCity;
signal input allowedRegion;
signal input minBirthDate;

signal input electionId;

// Outputs
signal output nullifier;
```

- Správa průzkumu
- Oddělení verifikační logiky
- Filtrace účastníků pomocí parametrů
- Validace hlasu a kontrola duplicit

```
function vote(
    uint256[2] memory a,
    uint256[2][2] memory b,
    uint256[2] memory c,
    uint256[10] memory input,
    uint256 _voteChoice
) public {
    require(block.timestamp < electionEndTime, "Error: Election has ended!");
    require(_voteChoice > 0 && _voteChoice <= numCandidates, "Error: Invalid candidate ID!");

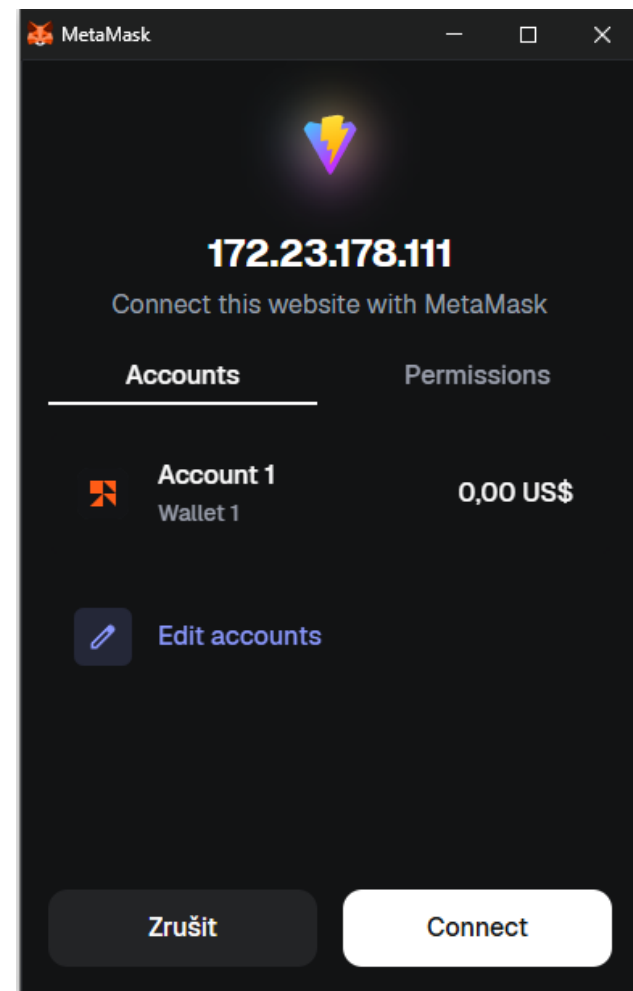
    uint256 proofNullifier = input[0];
    uint256 proofCity = input[5];
    uint256 proofRegion = input[6];
    uint256 proofMinDob = input[7];
    uint256 proofElectionId = input[9];

    require(proofCity == ALLOWED_CITY, "Error: Invalid City in proof");
    require(proofRegion == ALLOWED_REGION, "Error: Invalid Region in proof");
    require(proofMinDob == MIN_BIRTH_DATE, "Error: Invalid Birth Date in proof");
    require(proofElectionId == ELECTION_ID, "Error: Proof is for a different election!");

    require(!nullifiers[proofNullifier], "Error: User already voted!");
    require(verifier.verifyProof(a, b, c, input), "Error: Invalid ZK Proof!");

    nullifiers[proofNullifier] = true;
    candidateVotes[_voteChoice] += 1;
}
```

- Vývoj v Reactu
- Integrace s blockchainem přes MetaMask



Deploy Contract

Select Network

Sapphire Testnet

Sapphire Mainnet

Region

South Moravia

City

Brno

Duration (mins)

10

Candidates

4

Min Birth Date

01/22/2008

Deploy

✓ Election Deployed!

CONTRACT ADDRESS

0xECB3412e91Db278DEFD389B0FB77d2051A627117

🕒 Election Ends: 22. 1. 2026 20:08:15

📄 ZK PROOF CONFIGURATION

Copy this for the voting client

```
{
  "contractAddress": "0xECB3412e91Db278DEFD389B0FB77d2051A627117",
  "electionId": "1769108272",
  "allowedCity": "Brno",
  "allowedRegion": "South Moravia",
  "minBirthDate": "2008-01-22",
  "_intFormat": {
    "city": "003032593",
    "region": "116550661",
    "minDob": "20080122"
  }
}
```

Done

Select Network

Sapphire Testnet

Sapphire Mainnet

Cast Your Vote

Private voting using Zero-Knowledge Proofs on Oasis Sapphire.

STEP 1: GENERATE PROOF

Select Identity

Petr Svoboda (ID: 102) - Brno

Election ID

1769111626

Region

South Moravia

City

Brno

Min Birth Date

01/22/2008

Generate zk-SNARK Proof

STEP 2: SUBMIT TO BLOCKCHAIN

Contract Address

0x5d16f6881049Dab7Ce7D620003cBF23eAcE1709C

Candidate ID (Index)

1

Cast Vote

Election Results

Select Network

Sapphire Testnet

Sapphire Mainnet

0x5d16f6881049Dab7Ce7D620003cBF23eAcE1709C

Search

VOTING IN PROGRESS

22. 1. 2026 21:23:49

ELECTION ID

1769111626



Results are Encrypted
This election is currently active. The vote tally is hidden on-chain. Results will be revealed automatically when the election ends.

0x5d16f6881049Dab7Ce7D620003cBF23eAcE1709C

Search

ELECTION ENDED

22. 1. 2026 21:23:49

ELECTION ID

1769111626

Total Votes Cast: 3

Candidate #1

67%
2 votes

Candidate #2

33%
1 votes

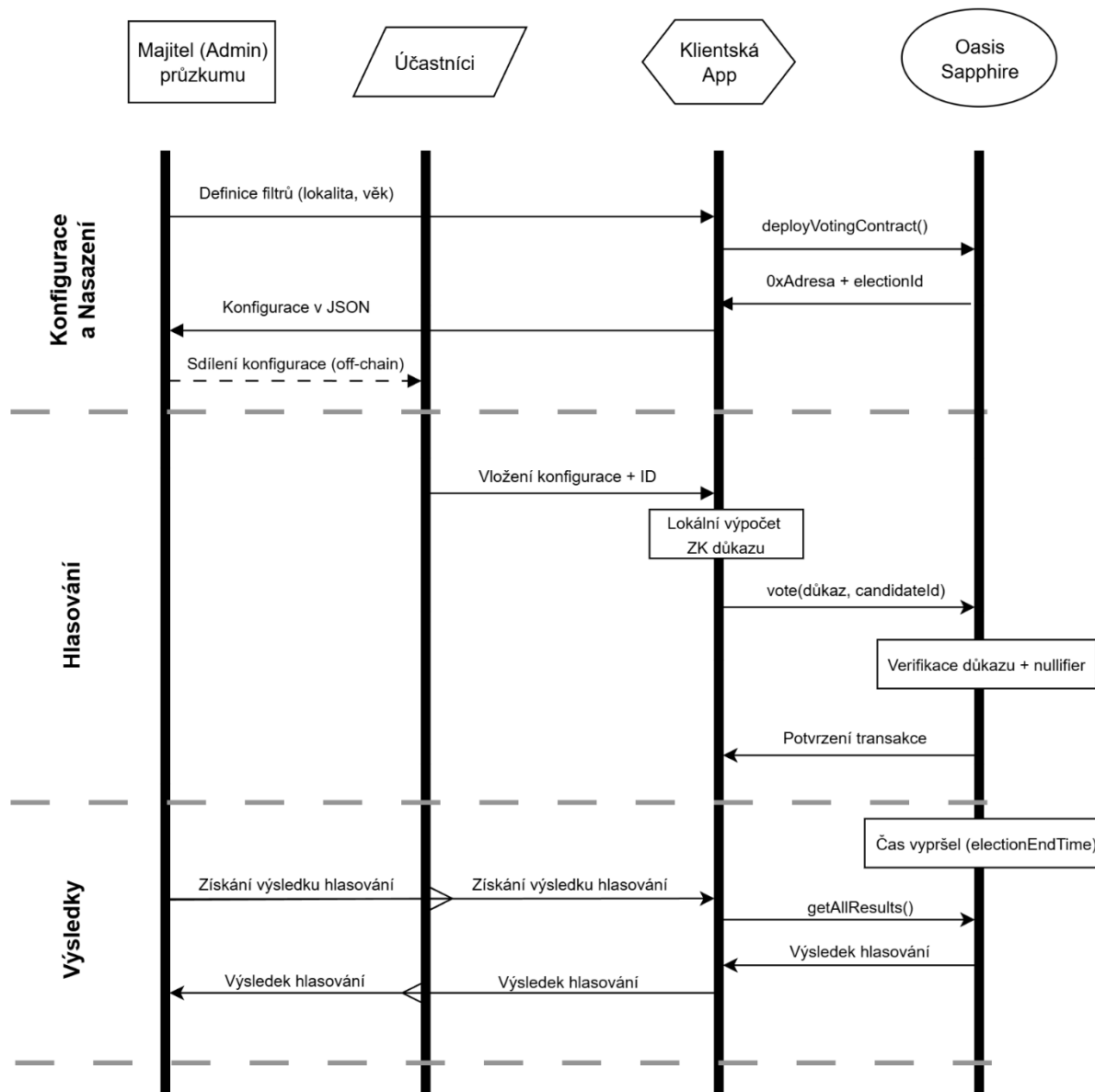
Candidate #3

0%
0 votes

Candidate #4

0%
0 votes

Transactions										Events	Transfers	Tokens (ERC-20)	NFTs (ERC-721)	Code
Filter by type Any														
Status	Encryption	Hash	Block	Age	Type	From	To	Amount	Fee					
✓	🔒	0xc940...7dcadb	15,387,720	45 min	📄 Contract Call	0x1F83...316367 →	0x5d16...E1709C		0.04 TEST					
✓	🔒	0x56ed...db771e	15,387,714	45 min	📄 Contract Call	0x1F83...316367 →	0x5d16...E1709C		0.03 TEST					
✓	🔒	0xe649...05d4c8	15,387,708	46 min	📄 Contract Call	0x1F83...316367 →	0x5d16...E1709C		0.04 TEST					
✓	🔒	0xeaa8...7e1d4d	15,387,585	58 min	📄 Contract Creation	0x1F83...316367 →	0x5d16...E1709C		0.14 TEST					



- Budoucí integrace EUDI
- Motivační mechanismy
- Rozšíření logiky a demografických filtrů

DĚKUJI ZA POZORNOST!